Silicon A/B Testing for Emergency Alerts

Rapid Pre-Deployment Stress Tests with LLM Resident Agents

Anonymous Author(s)

Abstract

Emergency alerts must communicate actionable guidance quickly under stress, yet small differences in message structure and wording can change what recipients understand and remember. Rigorous pre-deployment testing of alert variants is often slow and expensive, limiting iteration by agencies. We present a lightweight A/B simulation framework that uses instruction-tuned LLMs as stratified "resident" respondents to stress-test emergency-alert rewrites. For each alert, we generate control and three variants (action-first, plain-language, constraint-aware) and evaluate them across resident profiles varying in English proficiency, mobility constraints, and trust in officials. Agents return structured JSON outputs, enabling automatic scoring of action recall, confusion, and intended compliance against human-specified required actions. Across 17 alerts, paired within-alert comparisons show that action-first formatting yields a small but consistent recall lift that remains stable when scaling from 40 to 80 agents, while subgroup analyses reveal heterogeneous effects. We also introduce an interpretable structure score that helps explain when formatting changes translate into recall gains. The framework is intended for rapid screening and failure-mode discovery, complementing—not replacing—human evaluation.

Keywords

A/B testing, emergency alerts, public warnings, risk communication, message design, large language models, simulation, pre-deployment evaluation, human-in-the-loop, equity analysis, failure-mode discovery

1 Introduction

Public alerts and warning messages must communicate actionable guidance quickly under stress, yet small design choices (e.g., order of instructions, formatting, and inclusion of key details) can change what people understand and remember [7, 9, 17]. In practice, rigorous testing of alternative alert wordings is expensive and slow, which limits rapid iteration before deployment [10, 14]. Recent work suggests large language models (LLMs) can be used as *simulated respondents* for early-stage evaluation, revealing aggregate patterns while not replacing human validation [1, 2, 15].

We introduce a lightweight A/B simulation framework for emergency-alert rewrites. For each alert, we generate multiple variants (action-first, plain-language, constraint-aware) and evaluate them against a stratified set of simulated residents varying in English proficiency, mobility, and trust. Agents produce structured JSON responses that support scoring action recall, confusion, and intended compliance, enabling paired within-alert comparisons. Our goal is rapid pre-deployment screening: identifying promising rewrites and potential regressions to prioritize subsequent human testing.

Contributions. We contribute: (1) a rapid pre-deployment stress-test harness for comparing alert rewrites under controlled prompts; (2) *stratified resident agents* as an equity-oriented probe to surface potential disparate impacts and regressions across slices; (3) a failure-mode view over structured outputs that supports human emergency managers iterating on message content (what to fix, not just a score); and (4) an interpretable structure score (EAS) that helps explain when formatting changes translate into recall gains.

Intended use. The harness is designed for emergency managers during drafting/approval to triage candidate rewrites and prioritize what should advance to stakeholder review and human evaluation.

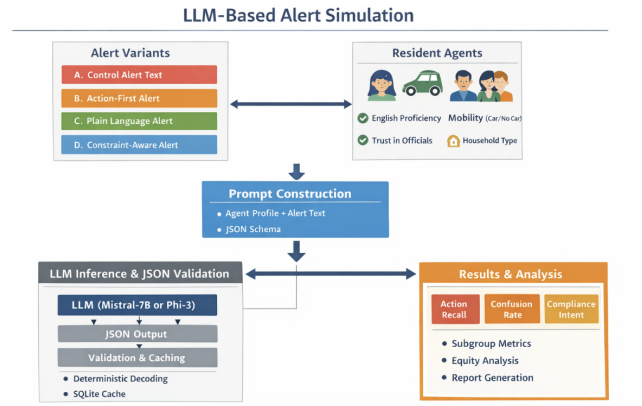

Figure 1: Simulation harness architecture. Alerts are transformed into variants (control, action-first, plain-language, constraint-aware) and evaluated against stratified resident profiles. Each trial returns structured JSON outputs that are scored against human-specified required actions and aggregated for paired A/B comparisons.

2 Related Work

Research on emergency warning communication emphasizes that message clarity, specificity, credibility cues, and actionability influence protective action, and that effects vary with recipient constraints and context [7, 9, 17]. Empirical studies in public alerting further examine how length and structure affect understanding and response; for example, work on tsunami and WEA messaging reports that constrained short formats can reduce comprehension, while longer messages can support improved outcomes when the extra space adds actionable detail [4, 16]. Systematic reviews of mobile/public alerting highlight persistent challenges around clarity, accessibility, and evaluation practices, motivating scalable methods for pre-deployment testing across diverse populations and hazards [10, 14].

Separately, recent LLM research explores using models as simulated respondents or agents for rapid experimentation and stress testing [1, 13, 15]. Because LLM outputs can encode biases and do not constitute behavioral ground truth, these approaches are best

Attribute	Levels	Example values
English proficiency	3	low / medium / high
Mobility access	2	constrained / unconstrained
Trust in officials	3	low / medium / high

Table 1: Resident stratification used as equity-oriented probes.

treated as screening tools that complement, rather than replace, human-subject evaluation [2, 3].

3 Methods

To ground our stress tests in realistic public-alert phrasing, we constructed a small corpus of $N = 17$ short alerts spanning weather, coastal/marine hazards, and public health advisories. Alert wording and terminology were informed by publicly available guidance and standards used in U.S. alerting practice. In particular, we drew on National Weather Service instructional directives describing standardized warning/advisory terminology and public-facing actions [11], the OASIS Common Alerting Protocol (CAP) specification that defines a widely used structure for emergency alert messages [12], and EPA Public Notification Rule guidance for standardized health-risk language in water system notifications [5]. Each alert includes a short list of human-specified *required actions* (e.g., "boil water", "avoid travel") used to score action recall under each variant.

Alert variants. For each alert text, we generate four variants: `A_control` (original), `B_action_first` (action steps moved to the top, formatted for scanability), `C_plain` (plain-language rewrite), and `D_constraint` (constraint-aware addendum that appends feasibility/support cues for common barriers). We then run a pre-deployment A/B-style stress test over the alert set (Figure 1).

Stratified resident agents (equity-oriented probes). We evaluate each variant against a stratified population of simulated residents defined by (i) English proficiency, (ii) mobility access, and (iii) trust in officials (Table 1; 40-agent and 80-agent runs; other attributes such as time pressure and household type are held fixed or treated as metadata). We treat these profiles as *stress-test instruments* for surfacing potential disparate impacts and regressions across slices—not as demographic inference or behavioral ground truth. Accordingly, in addition to aggregate effects, we inspect slice-level paired deltas to identify variants that improve average recall while degrading performance for specific profiles (a "do-no-harm" screening criterion) prior to stakeholder review or human-subject evaluation.

LLM setup and trial procedure. We use instruction-tuned open-weight LLMs in a Google Colab GPU environment (4-bit quantization) with deterministic decoding (temperature = 0). Each (alert × agent × variant) trial prompts the model to output *JSON only* (keys: `recalled_actions`, `confusion`, `intended_compliance`). We compute action recall as the fraction of required actions present in the recalled-action list (with light normalization), and we record confusion and intended compliance as additional screening signals. Because using the same model family for generation and evaluation can introduce shared artifacts, we emphasize within-trial pairing and deterministic decoding; nevertheless, this does not eliminate evaluator–generator coupling, reinforcing our framing as rapid screening rather than validation.

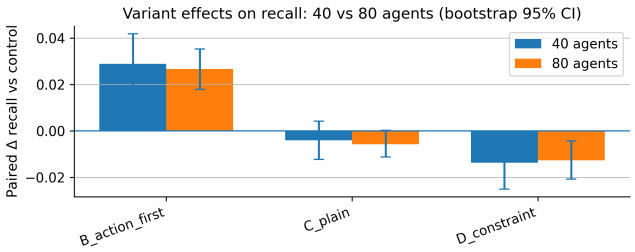

Figure 2: Paired Δ action recall vs. `A_control` (bootstrap 95% CI), comparing 40-agent and 80-agent runs.

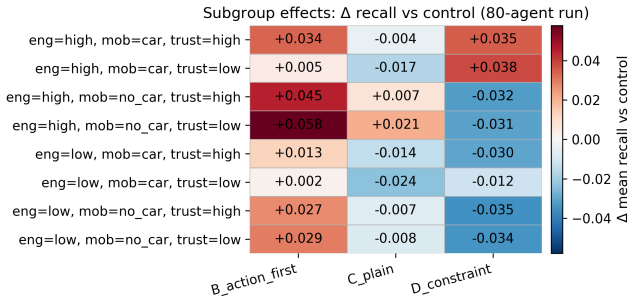

Figure 3: Subgroup heatmap (80-agent run): paired Δrecall vs. control by resident attribute slices.

Paired effects and uncertainty. To isolate messaging effects from alert difficulty and agent idiosyncrasies, we report paired deltas within each (alert × agent) pair: $\Delta = \text{metric(variant)} - \text{metric(control)}$. We summarize effects by variant and by resident slice, and estimate uncertainty using bootstrap confidence intervals.

4 Results

We report action recall as the fraction of required actions correctly recalled (0–1). To isolate messaging effects from alert/agent idiosyncrasies, we use *paired deltas* computed within each (alert × agent) trial: for each variant $v \in \{\texttt{B_action_first}, \texttt{C_plain}, \texttt{D_constraint}\}$, $\Delta\text{recall}_v = \text{recall}_v - \text{recall}_{\texttt{A_control}}$. This framing follows common practice in warning-message evaluation and helps connect our simulated experiments to empirical messaging work on mobile/public alerts [4, 10, 14, 16].

4.1 Paired A/B effects on recall are stable across agent counts

Figure 2 shows mean paired Δrecall with bootstrap 95% CIs for the 40-agent and 80-agent runs. The direction of effects is consistent across runs: `B_action_first` yields a small positive change in recall relative to control, while `C_plain` and `D_constraint` are near-zero to slightly negative on average. This pattern is aligned with the warning-design intuition that making action steps more immediately scannable can improve comprehension/recall [7, 9, 17].

4.2 Subgroup analysis highlights distributional variation

Aggregate effects mask heterogeneity across resident profiles. Figure 3 summarizes paired Δrecall across attribute slices (English proficiency, mobility, trust, time pressure, household). We observe non-uniform effects: some slices benefit more from `B_action_first`, while others show negligible or negative changes, suggesting that "one-size-fits-all" rewrites may under-serve certain populations. This mirrors long-standing findings that warning effectiveness depends on recipient constraints and context (e.g., resource limits, ability to act, perceived credibility) [7, 9, 14].

4.3 Derived failure modes suggest where variants trade off performance

To move beyond a single scalar outcome, we derive coarse failure modes from the structured outputs (e.g., missing actions vs. partial recall vs. confusion markers). Figure 4 shows that the dominant failure is *missing required actions*, with a substantial secondary mass in *partial recall*. Across variants, the main differences appear as shifts between these categories rather than elimination of failures. This is practically relevant for alerting systems: small average changes may still correspond to meaningful shifts in *which* action steps are omitted [9, 17].

HITL iteration via actionable failure-mode diagnostics. We frame failure modes as design feedback for emergency managers: "missing actions" suggests adding or front-loading omitted required steps; "partial recall" suggests reducing competing instructions and using numbered action slots; and "confusion-linked low recall" suggests removing jargon and clarifying time windows (NOW/UNTIL) and affected populations. This supports an iterative loop: draft → simulate → inspect failure-mode shifts → revise → re-test, before any human-subject evaluation.

Worked example (message revision prompted by a failure mode). For instance, consider a boil-water alert whose required actions include *boil tap water for at least one minute* and *use bottled water if boiling is not possible*. In multiple trials under `C_plain`, agents frequently produced *partial recall*: they recalled "boil water" but omitted the minimum duration and the fallback option. A human reviser can respond by making the missing steps explicit and scannable in an action-first rewrite, e.g., "**DO NOW:** (1) Boil tap water for **1 minute** before drinking or cooking. (2) If you cannot boil water, use **bottled water**." Re-running the harness can then verify whether mass shifts from *partial recall* to *full recall* without increasing confusion, providing a targeted, auditable way to iterate on content rather than optimizing to a single score.

4.4 Message structure is associated with recall gains beyond grade-level readability

Finally, we test whether a simple *structure-focused* message score predicts recall. We define an Emergency Alert Structure (EAS) score $S(\cdot) \in [0, 100]$ from surface cues that support scanning and actionability (action placement, slot completeness, formatting, and length penalties), motivated by warning-design guidance and readability research [6, 8, 9, 17].

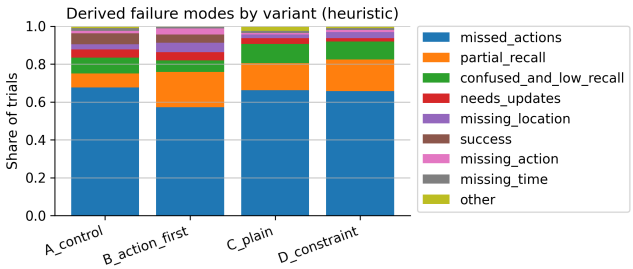

Figure 4: Derived failure modes by variant (heuristic from trial-level outputs): distribution over missing actions, partial recall, confusion-linked low recall, and other categories.

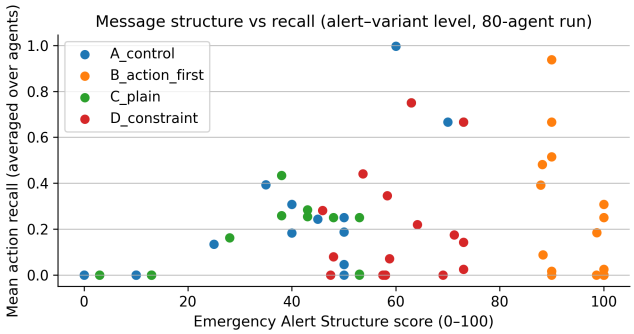

Figure 5: EAS score vs. mean recall (alert–variant level, 80-agent run).

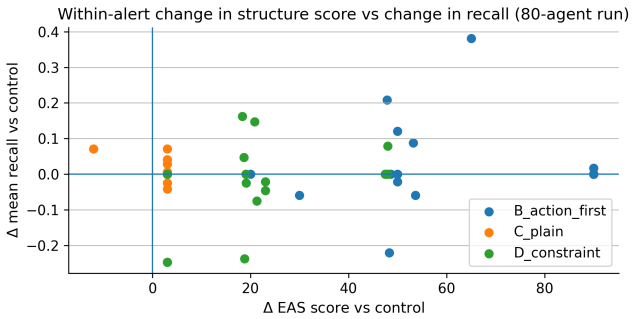

Figure 6: Within-alert ΔEAS vs. paired Δrecall relative to `A_control` (80-agent run), colored by variant.

Figure 5 shows that higher EAS is associated with higher mean recall at the alert–variant level. More importantly for causal interpretation within alert, Figure 6 plots within-alert changes: ΔS vs. paired Δrecall relative to `A_control`. The positive quadrant concentration for `B_action_first` suggests that improving scan-structure (action prominence + formatting) is a plausible mechanism for recall gains, consistent with prior WEA/message format arguments [4, 10].

In Summary. Across 17 alerts, `B_action_first` produces a small positive paired Δrecall that is directionally stable when scaling

from 40 to 80 agents (Figure 2), whereas `C_plain` and `D_constraint` are near-zero to slightly negative on average. However, subgroup slices show heterogeneous effects (Figure 3), motivating "do-no-harm" screening in which a candidate rewrite is flagged if it regresses for particular profiles. Failure-mode distributions are dominated by omitted required actions, with variant differences appearing primarily as shifts between missing-actions and partial-recall categories (Figure 4), which supports targeted human revision rather than optimization to a single score. Finally, a structure-focused score (EAS) aligns with recall at the alert–variant level and within-alert ΔEAS tracks Δrecall most clearly for `B_action_first` (Figures 5–6), suggesting scan-structure as a plausible mechanism.

5 Limitations

Our results are *screening signals*, not behavioral ground truth. LLM outputs can reflect model biases and may not capture real emotion, risk perception, or constraints on taking action [2, 3]. A key risk is *hallucinated compliance*: agents may report intent to comply even when real recipients face barriers (e.g., disability access, transport, childcare, competing obligations), which can inflate apparent effectiveness. Resident profiles are stylized and can invite "optimizing to the simulator" (rewrites that fit prompt artifacts rather than improving real comprehension). Accordingly, we treat slice-level outputs as *discovery* tools to prioritize stakeholder review and targeted human studies, not as evidence of fairness or real-world effectiveness. Finally, failure-mode labels are heuristic and depend on prompting and normalization choices.

6 Conclusion

We present an auditable, lightweight simulation harness for rapid A/B testing of emergency-alert rewrites. Across 17 alerts, action-first formatting yields a small, robust recall lift (stable from 40 to 80 agents), while subgroup analysis highlights distributional variation that motivates "do-no-harm" selection criteria. The framework is intended to accelerate early iteration and prioritize candidates for subsequent human-centered evaluation.

References

[1] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis* 31, 3 (July 2023), 337–351. doi:10.1017/pan.2023.2 Published online 21 Feb 2023.

[2] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*.

[3] Rishi Bommasani et al. 2021. On the Opportunities and Risks of Foundation Models. arXiv:2108.07258.

[4] Elizabeth J. Carlson, Hamilton Bean, Chelsea Ratcliff, Manu Pokharel, and Joshua Barbour. 2024. Do 360-Character Wireless Emergency Alert (WEA) Messages Work Better than 90-Character Messages? Testing the Risk Communication Consensus. e12587 pages. doi:10.1111/1468-5973.12587 Epub 2024-06-16. Open access via PubMed Central (PMCID: PMC11424238)..

[5] Environmental Protection Agency. 2000. *The Public Notification Rule: 40 CFR Part 141, Subpart Q*. United States Office of the Federal Register. https://www.epa.gov/dwreginfo/public-notification-rule Standardized health-risk language for water system contaminants..

[6] Rudolf Flesch. 1948. A New Readability Yardstick. *Journal of Applied Psychology* 32, 3 (1948), 221–233.

[7] Michael K. Lindell and Ronald W. Perry. 2012. The Protective Action Decision Model: Theoretical Modifications and Additional Evidence. *Risk Analysis* 32, 4 (2012), 616–632.

[8] G. Harry McLaughlin. 1969. SMOG Grading—A New Readability Formula. *Journal of Reading* 12, 8 (1969), 639–646.

[9] Dennis S. Mileti and John H. Sorensen. 1990. *Communication of Emergency Public Warnings: A Social Science Perspective and State-of-the-Art Assessment*. Technical Report ORNL-6609. Oak Ridge National Laboratory.

[10] Fiona Mowbray et al. 2024. A systematic review of the use of mobile alerting to inform the public. *Journal of Contingencies and Crisis Management* (2024). doi:10.1111/1468-5973.12499

[11] National Weather Service. 2023. *National Weather Service Instruction 10-511: WFO Post-Event Operations*. National Oceanic and Atmospheric Administration (NOAA). https://www.nws.noaa.gov/directives/ Defines standardized warning and advisory terminology and required public actions..

[12] OASIS Standard. 2010. *Common Alerting Protocol Version 1.2*. Technical Report. Organization for the Advancement of Structured Information Standards. http://docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2-os.html

[13] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. Also available as arXiv:2304.03442..

[14] Abdul-Akeem Sadiq, Ratna B. Dougherty, Jenna Tyler, and Rebecca Entress. 2023. Public alert and warning system literature review in the USA: identifying research gaps and lessons for practice. *Natural Hazards* 117, 2 (2023), 1711–1744. doi:10.1007/s11069-023-05926-x Published online 11 April 2023.

[15] Shicheng Sun et al. 2024. Random Silicon Sampling: Simulating Human Sub-Population Opinion Using a Large Language Model Based on Group-Level Demographic Information. arXiv:2402.18144.

[16] Jeannette Sutton, Sarah C. Vos, Michelle M. Wood, and Matthew Turner. 2018. Designing Effective Tsunami Messages: Examining the Role of Short Messages and Fear in Warning Response. *Weather, Climate, and Society* 10, 1 (2018), 75–87. doi:10.1175/WCAS-D-17-0032.1

[17] Michael S. Wogalter (Ed.). 2006. *Handbook of Warnings*. CRC Press.